Article

# A Context-Based Perspective for Frost Analysis in Reuse-Oriented Big Data System Developments

**Agustina Buccella [1,†,‡] 0000-0002-8516-7453, Alejandra Cechich [1,‡] 0000-0003-4804-6270, Federico Saurin [1,‡], Ayelén Montenegro [2,‡], Andrea Rodríguez [2,‡] and Angel Muñoz [2,*]**

[1] GIISCO Research Group, Departamento de Ingeniería de Sistemas, Facultad de Informática, Universidad Nacional del Comahue, Neuquen, Argentina; agustina.buccella,alejandra.cechich,federico.saurin@fi.uncoma.edu.ar
[2] Instituto Nacional de Tecnología Agropecuaria (INTA), Alto Valle de Río Negro y Neuquén, Argentina; montenegro.ayelen,rodriguez.andrea,munoz.angel@inta.gob.ar
* Correspondence: agustina.buccella@fi.uncoma.edu.ar; Tel.:
† Current address: Affiliation.
‡ These authors contributed equally to this work.

**Abstract:** The large amount of available data, generated every second by sensors, social networks, organizations, and so on, have generated new lines of research that involve novel methods, techniques, resources and/or technologies. The development of Big Data Systems (BDS) can be approached from different perspectives, all of them useful depending on the objectives pursued. In particular, in this work we address BDS from the area of software engineering, contributing to the generation of novel methodologies and techniques towards software reuse. In this article, we propose a methodology to develop reusable BDS by mirroring activities from software product line engineering. It means that the process to build BDS is approached by analyzing *Variety* of domain features and modeling them as a family of related assets. The contextual perspective of the proposal, along with its supporting tool, are introduced through a case study in the agrometeorology domain. The characterization of variables for frost analysis exemplifies the importance of identifying variety, as well as the possibility of reusing previous analyses adjusted to the profile of each case. In addition to show interesting findings from the case, we also exemplify our concept of *context variety*, which is a core element to model reusable BDS.

**Keywords:** Reusability; Big Data Systems; Variety identification; Agrometeorology domain

## 1. Introduction

Success in global markets for software-intensive products depends, among other things, on shortening the time to market but improving quality and evolution or reducing the resources required. This aspect can be addressed through external or internal strategies. External strategies include acquiring off-the-shelf components and having development, maintenance, and support performed by third parties. In-house strategies include global software development, where resources are geographically distributed according to specific needs, profiles, conditions, or costs [1], and software product line (SPL) engineering and ecosystems, i.e., *the strategic acquisition, creation, and reuse of software assets* [2–4]. Our research focuses on the latter context – that of engineering product lines and ecosystems. This engineering differs from traditional systems development in two fundamental respects [5]:

- It requires two distinct development processes: domain engineering and application engineering. The first defines and models the commonality and variability, establishing a platform to rapidly develop quality applications within the line. Application engineering derives specific applications by strategically reusing that platform and exploiting its variability.
- It needs to explicitly define and manage variability. During domain engineering, variability is introduced into all software assets such as domain requirements, architectural models, components, or test cases. During application engineering, it is exploited to derive large-scale customized applications according to the needs of customers and markets.

**Citation:** Buccella, A.; Cechich, A.; Saurin, F; Montenegro, A.; Rodríguez, A; Muñoz, A. A Context-Based Perspective for Frost Analysis in Reuse-Oriented Big Data System Developments. *Information* **2024**, *1*, 0. https://doi.org/

Received:
Revised:
Accepted:
Published:

information
an Open Access Journal by MDPI

CITESCORE 6.9
IMPACT FACTOR 2.4

# CERTIFICATE OF ACCEPTANCE

The certificate of acceptance for the manuscript (**information-3205119**) titled:

A Context-Based Perspective for Frost Analysis in Reuse-Oriented Big Data System Developments

Authored by:

Agustina Buccella; Alejandra Cechich; Federico Saurin; Ayelén Montenegro; Andrea Rodríguez; Angel Muñoz

was accepted in *Information* (ISSN 2078-2489) on 17 October 2024

MDPI Academic Open Access Publishing since 1996

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# [Information] Manuscript ID: information-3205119 - Accepted for Publication

1 mensaje

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Manuscript ID: information-3205119
Type of manuscript: Article
Title: A Context-Based Perspective for Frost Analysis in Reuse-Oriented Big Data System Developments
Authors: Agustina Buccella *, Alejandra Cechich, Federico Saurin, Ayelén Montenegro, Andrea Rodríguez, Angel Muñoz
Received: 28 Aug 2024
E-mails: agustina.buccella@fi.uncoma.edu.ar, alejandra.cechich@fi.uncoma.edu.ar, federico.saurin@est.fi.uncoma.edu.ar, montenegro.ayelen@inta.gob.ar, rodriguez.andrea@inta.gob.ar, munoz.angel@inta.gob.ar
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